Primary and Secondary Amenorrhea

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Learning Objectives

- Define primary amenorrhea and recognize its clinical presentation
- Recognize the differential diagnosis of primary amenorrhea
- Describe the work-up and treatment considerations for primary amenorrhea
- Define secondary amenorrhea and recognize its clinical presentation
- Recognize the differential diagnosis of secondary amenorrhea
- Describe the work-up and treatment of secondary amenorrhea

Primary Amenorrhea

Primary Amenorrhea: Definition and Clinical Presentation

- No periods by age 15y in the presence of normal growth and secondary sexual development

- When should we initiate workup?
  - No period by age 15y
  - No period PLUS no breast development by age 13y
  - No period within 5y of onset of breast development
  - No period by age 14y with clinical evidence of hyperandrogenism (hirsutism, cystic acne)

Primary Amenorrhea: Differential Diagnosis

Structural/outflow tract abnormalities
- 5a-reductase deficiency
- Androgen insensitivity syndrome
- Mullerian agenesis
- Imperforate hymen
- Transverse vaginal septum

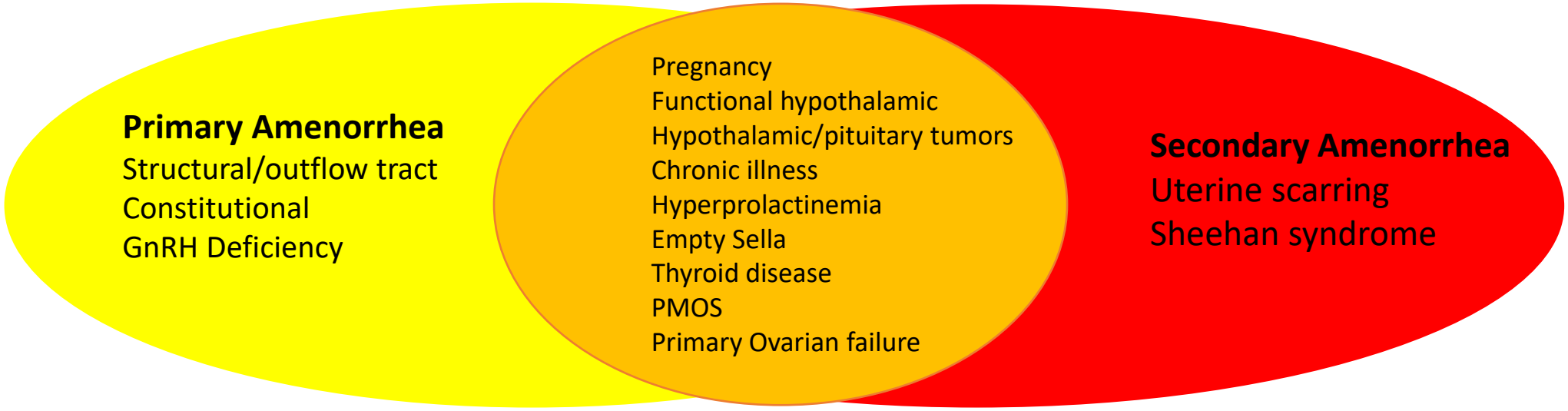
Primary ovarian insufficiency
- Turner syndrome (45X)
- Other gonadal dysgenesis (46XX or 46XY)
- Idiopathic
- Acquired (chemo or radiation exposure, surgical)

Hypothalamic or pituitary disorders
- GnRH deficiency/defect
- Constitutional delay

Primary Amenorrhea: Differential Diagnosis

- The secondary amenorrhea causes!
  - Thyroid, pituitary, functional (stress, low BMI, high exercise, chronic illness), Polyendocrine Metabolic Ovarian Syndrome (PMOS, previously PCOS)
  - PREGNANCY!

  - We will get into these in the second half of the lecture



Structural/Outflow Tract: 5a-Reductase Deficiency

5a-reductase deficiency

- 46,XY with normal testosterone production but defective conversion of testosterone to dihydrotestosterone (DHT) --> impaired virilization during embryogenesis
- 5-alpha-reductase type 2 enzyme is defective (various **autosomal recessive** loss of function mutations) but type 1 enzyme is normal
- Patients may have low (but measurable) serum DHT levels due either to residual activity of the defective enzyme or the normal type 1 isoenzyme
- Virilization at the time of expected puberty may occur due to increase in serum testosterone or increase in DHT concentration through the Type 1 enzyme
- Testosterone and estrogen production are normal/expected range for an XY individual, which is why gynecomastia does not develop (as opposed to androgen insensitivity- see next slide).
- Genital development
  - External genitalia varies from isolated microphallus with varying degrees of hypospadias (33%), to typical appearing labia and clitoris (7%), labia with clitoromegaly (49%)
  - The internal urogenital tract consists of epididymis, vas deferens, seminal vesicles, and ejaculatory ducts. These empty into a blind-ending vagina (55%), on the perineum near the urethra (40%), or at the end of the phallus (5%). The testes are outside the abdominal cavity in the inguinal canals, labia majora, or scrotum. Spermatogenesis is most often impaired.

Structural/Outflow Tract: Androgen Insensitivity Syndrome (AIS)

Complete Androgen Insensitivity Syndrome

- 46 XY with loss of function mutations in the gene that codes for the androgen receptor
- Complete AIS is the 3rd most common cause of primary amenorrhea
  - Partial AIS has a variable phenotype
- Little or no axillary or pubic hair
- In infancy, inguinal mass may be undescended testicle
  - Testes may be in the abdomen, inguinal canals or labia majora
- Presence of labia and clitoris, absent uterus, short vagina
- Elevated LH due to resistance to feedback of androgens at the hypothalamic-pituitary level --> increased testosterone levels
  - Breast development occurs (excess testosterone aromatizes to estrogen)

Structural/Outflow Tract: Mullerian Agenesis

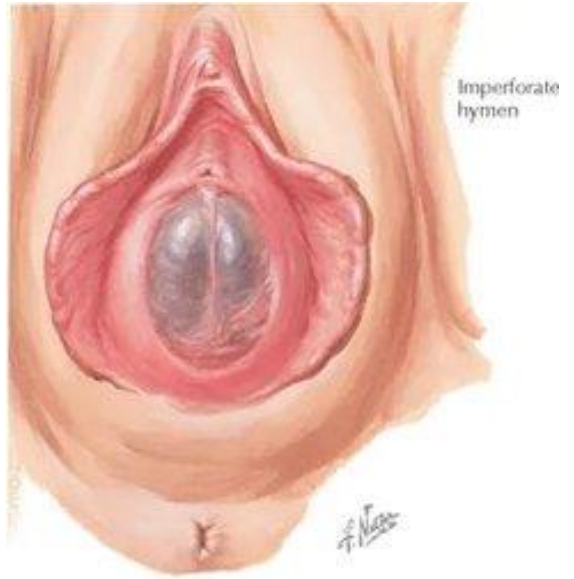
Mullerian agenesis (aka Vaginal agenesis, Mayer-Rokitansky-Kuster-Hauser syndrome, MRKH)

- 46 XX karyotype with agenesis or hypoplasia of mullerian duct system
  - Underlying etiology largely unknown/unidentified
- Congenital absence of the vagina most often accompanied by cervical and uterine agenesis (<10% have an obstructed or rudimentary uterus)
- 25-50% have urologic abnormalities (horseshoe kidney, unilateral renal agenesis)
- 10-15% have skeletal abnormalities
- Normal ovaries and normal ovarian function, thus typical secondary sexual differentiation EXCEPT no menarche (primary amenorrhea)

Structural/Outflow Tract: Vagina

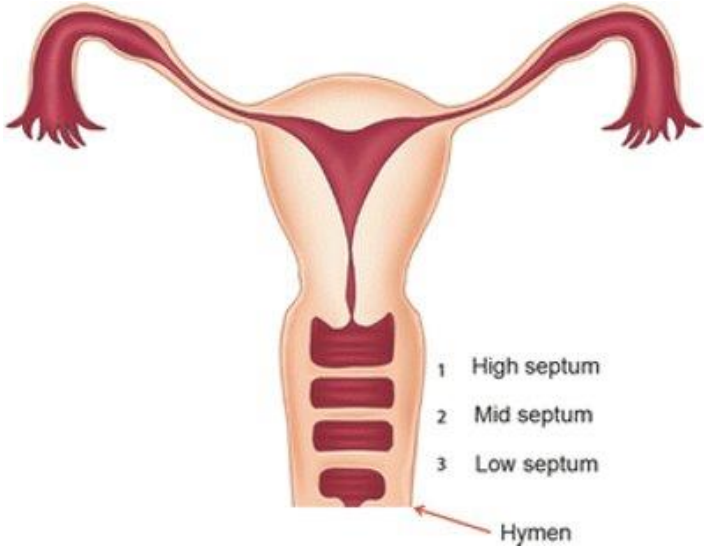
Imperforate hymen

- Hymen is ALWAYS at the vaginal opening, so an imperforate hymen would not be high in the vagina
- Hymen is thin: 1-2mm
    - Can appear blue hued when blood trapped behind (hematocolpos)

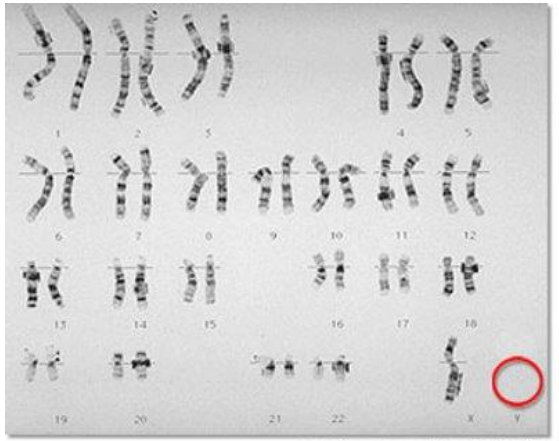


Transverse vaginal septum

- Transverse vaginal septum <u>most commonly</u> 2/3 of the way up the vagina (high septum)
    - Can also be mid or low
- Septum is thick: 1-2cm

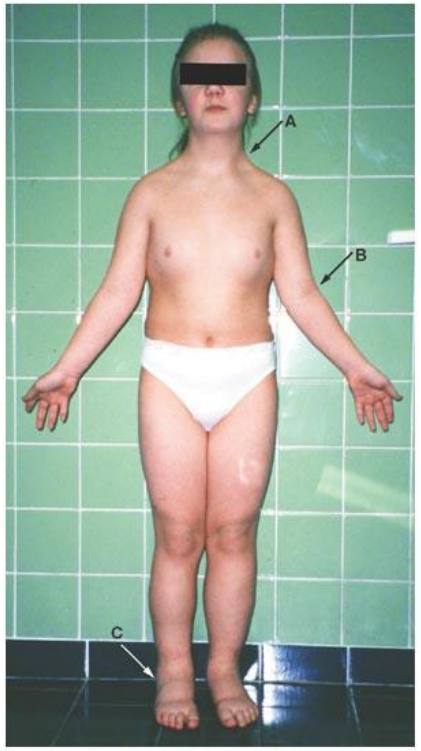


Primary Ovarian Insufficiency



Turner syndrome-

- 45X gonadal dysgenesis- missing an entire X chromosome
  - Can also have mosaic 45X/46XX- spontaneous puberty and menstruation typically occur, followed by secondary amenorrhea
- Premature depletion of oocytes/follicles (largely in utero) --> decreased estradiol and inhibin B --> significantly elevated FSH due to reduced negative feedback
- Ovaries are replaced with fibrous tissue
  - Should have estradiol and progesterone therapy
- Other typical features: short stature, shield chest/widely spaced nipples, short and webbed neck. Hearing loss, hypothyroidism and LFT abnormalities are common with increasing age. Increased risk of cardiovascular disease due to cardiac malformations.



Primary Ovarian Insufficiency

Other gonadal dysgenesis (hypogonadism)

- 46XX gonadal dysgenesis
  - Rare, 46XX but has underdeveloped non-functional "streak ovaries"
  - Many genetic causes discovered with next generation sequencing
  - Most often mutations in genes that code for proteins involved in homologous recombination in meiosis and DNA damage/repair
- 46XY gonadal dysgenesis (Swyer Syndrome)
  - Fibrous streak gonad doesn’t secrete anti-mullerian hormone (AMH) resulting in persistent mullerian structures (vulva, vagina, uterus and fallopian tubes)
  - Various mutations have been identified, including mutations in or absence of SRY
  - Can be inherited or de novo

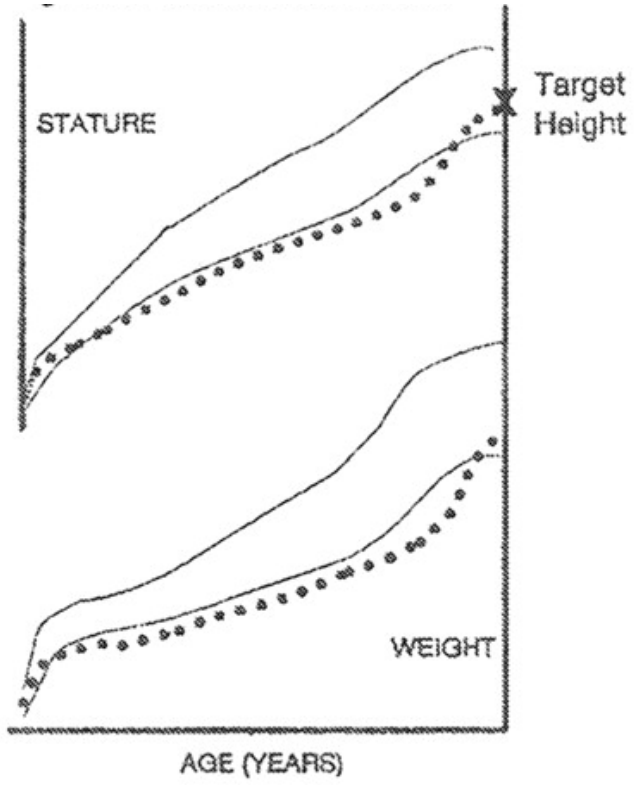
Hypothalamic and Pituitary Disorders

GnRH deficiency (isolated GnRH deficiency, idiopathic hypogonadotropic hypogonadism)

- Defects in production and/or action of GnRH
  - About 1/3 are inherited (autosomal dominant, autosomal recessive, or X-linked recessive inheritance)- many different genes are implicated in etiology
  - The remaining 2/3 are sporadic
- Can occur with normal olfaction or with anosmia (Kallman syndrome)
- Inappropriately low or low normal FSH and LH in the setting of prepubertal gonadal steroid levels
- Difficult to directly diagnose with GnRH levels so diagnosis inferred by
  - Lack of LH pulses in frequent blood sampling
  - Robust gonadotrophin secretion in response to pulsatile GnRH administration demonstrating functional response of pituitary

Hypothalamic and Pituitary Disorders

- Constitutional Delay
  - Diagnosis of exclusion
  - Will go on to typical pubertal development, just later
  - Can be difficult to distinguish from GnRH deficiency
  - Genetic basis poorly understood



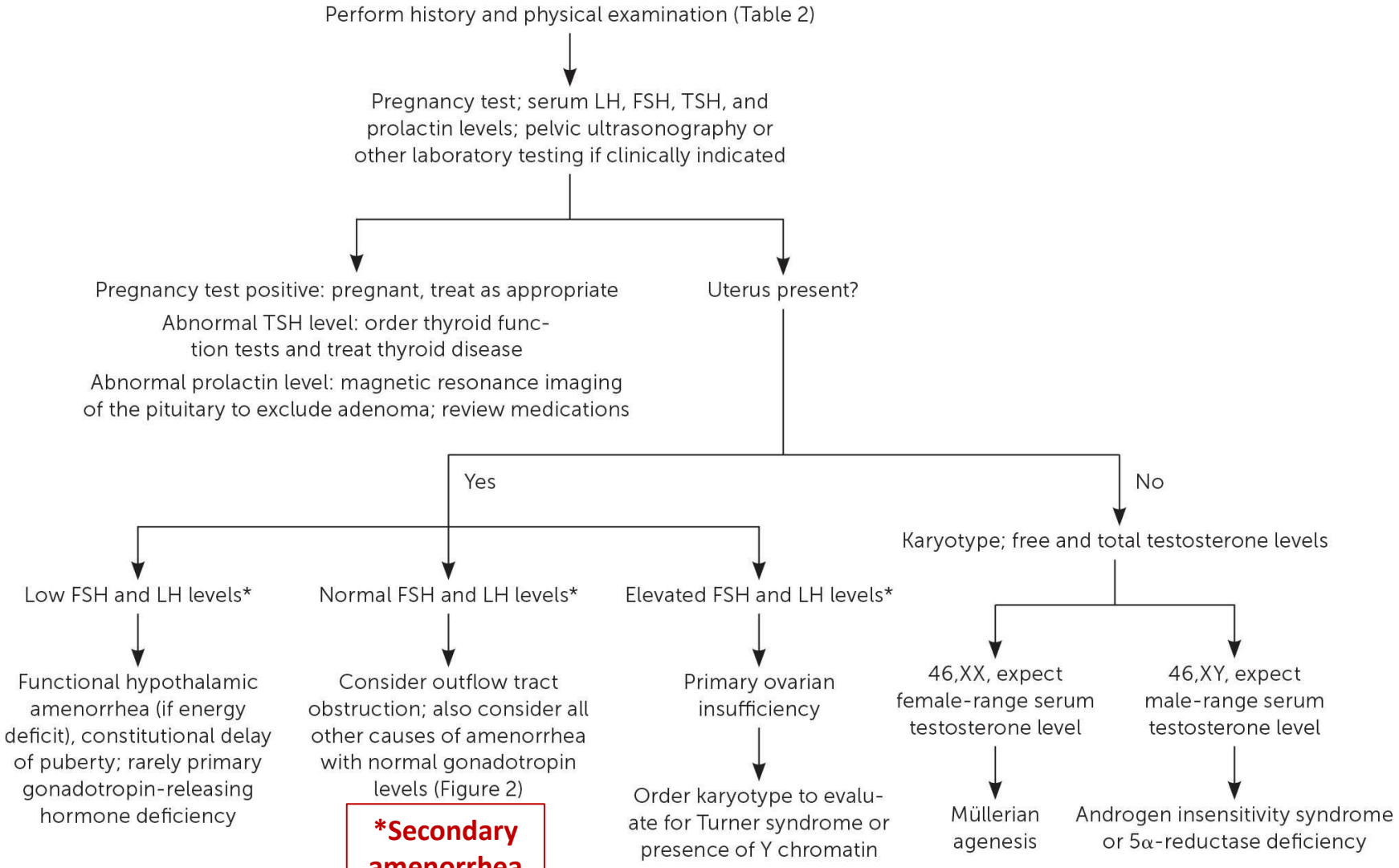
Primary Amenorrhea: H&P

Findings in the Evaluation of Amenorrhea

Findings	Associations
History	
Chemotherapy or radiation	Impairment of specific organ or structure, (e.g., brain, pituitary, ovary)
Family history of early or delayed menarche	Constitutional delay of puberty
Galactorrhea	Pituitary tumor
Hirsutism, acne	Hyperandrogenism, PCOS, ovarian or adrenal tumor, CAH, Cushing syndrome
Illicit or prescription drug use	Multiple associations, consider effect on prolactin
Loss of smell (anosmia)	Kallman syndrome (GnRH deficiency)
Menarche and menstrual history	Primary vs. secondary amenorrhea
Sexual activity	Pregnancy
Significant headaches or vision changes	Central nervous system tumor, empty sella syndrome
Temperature intolerance, palpitations, diarrhea, constipation, tremor, depression, skin changes	Thyroid disease
Vasomotor symptoms (e.g., hot flashes or night sweats)	Primary ovarian insufficiency, natural menopause
Weight loss, excessive exercise, poor nutrition, psychosocial distress, diets	Functional hypothalamic amenorrhea
Physical examination	
Abnormal thyroid examination	Thyroid disorder
Acanthosis nigricans or skin tags	Hyperinsulinemia (PCOS)
Anthropomorphic measurements; growth charts	Multiple associations; Turner syndrome, constitutional delay of puberty
Body mass index	High: PCOS
	Low: Functional hypothalamic amenorrhea
Bradycardia	Functional hypothalamic amenorrhea (e.g., anorexia nervosa)
Breast development (normal progression)	Presence of circulating estrogen*
Dysmorphic features (e.g., webbed neck, short stature, low hairline)	Turner syndrome
Male pattern baldness, increased facial hair, acne	Hyperandrogenism, PCOS, ovarian or adrenal tumor, CAH, Cushing syndrome
Pelvic examination	
Absence or abnormalities of cervix or uterus	Rare congenital causes including Müllerian agenesis or androgen insensitivity syndrome
Clitoromegaly	Androgen-secreting tumor; CAH; 5α-reductase deficiency
Presence of transverse septum or imperforate hymen	Outflow tract obstruction
Reddened or thin vaginal mucosa	Decreased endogenous estrogen
Sexual maturity rating abnormal	Turner syndrome, constitutional delay of puberty, rare causes
Striae, buffalo hump, central obesity, hypertension	Cushing syndrome

Amenorrhea: A Systematic Approach to Diagnosis and Management | AAFP

Primary Amenorrhea: Work-Up



***Secondary amenorrhea causes**

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Primary Amenorrhea: Treatment

- Diagnosis is the goal, not necessarily periods
  - Risk assessment (undescended testes, cardiovascular disease, etc)
  - Fertility goals/possibilities
  - Addressing social and developmental outcomes
- If due to outflow tract abnormality
  - Obstruction: Remove obstructing tissue
  - Absent uterus/vagina: Assess patient goals
- For secondary causes, treat the cause



Secondary Amenorrhea

Secondary Amenorrhea: Definition and Clinical Presentation

- The absence of menses for six or more months

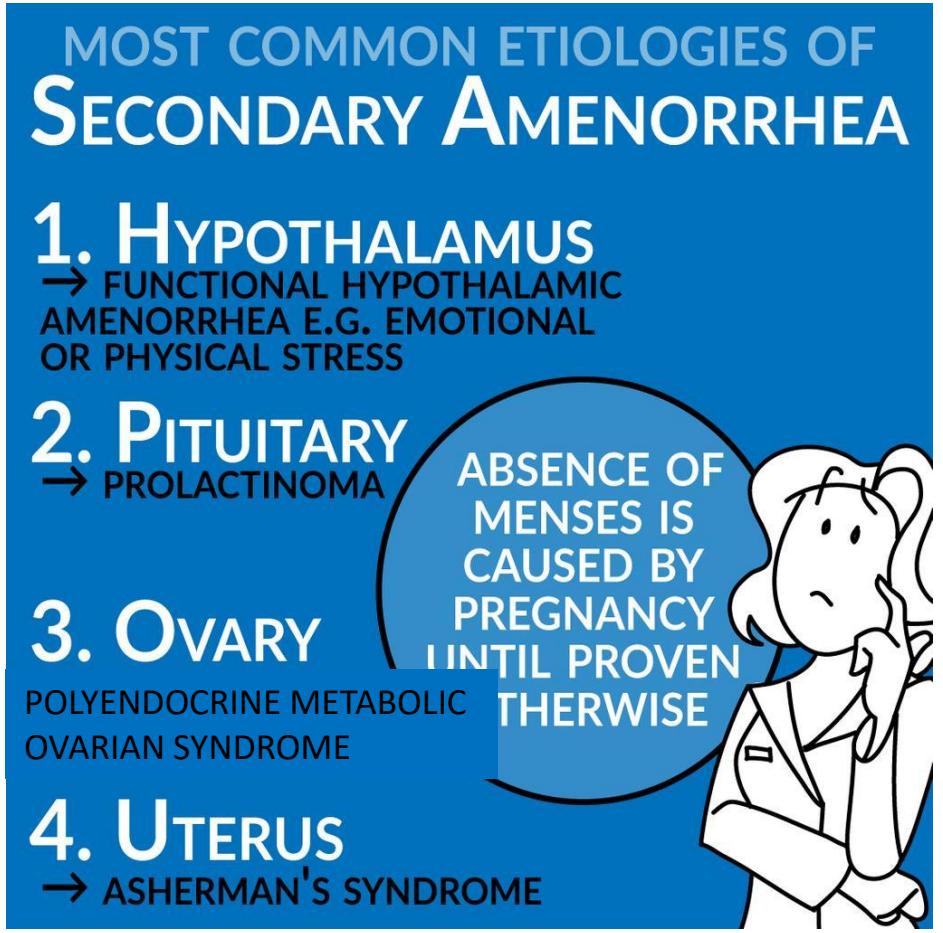
OR

- The absence of menses for the length of three cycles after the establishment of regular menstrual cycles (e.g. A person with regular 28-30 day cycles would meet this after 3 months)

Secondary Amenorrhea: Differential Diagnosis

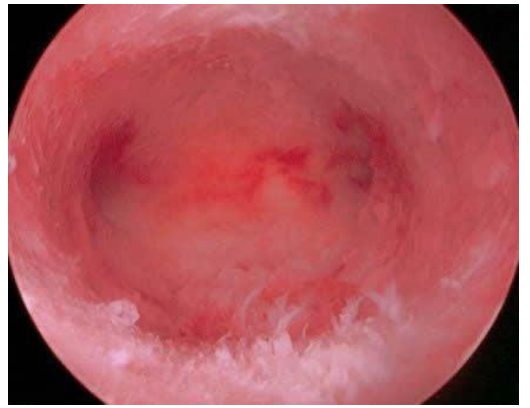
- Pregnancy

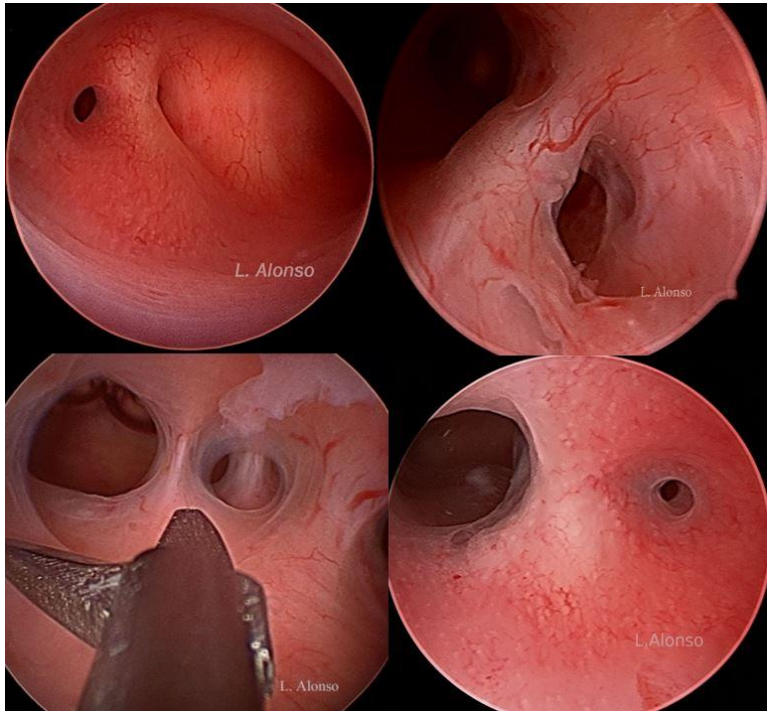
- Uterine
- Ovarian
- PMOS/Hyperandrogenism
- Thyroid
- Pituitary
- Hypothalamic



Secondary Amenorrhea: Uterine Cause

- Intrauterine adhesions (Asherman Syndrome)
  - Acquired scarring of endometrium
  - Usually secondary to postpartum hemorrhage management or D&C (dilation and curettage) after pregnancy loss or termination of pregnancy
  - Patients must have a preceding event after which their periods became absent or very light- does not happen spontaneously





Secondary Amenorrhea: Ovarian Causes

- Premature Ovarian Insufficiency/Failure
  - Depletion of oocytes under age 40y
  - Review: Low estradiol and inhibin B leads to loss of negative feedback on the hypothalamus/pituitary resulting in high FSH levels
  - Can be seen in mosaic 45X/46XX, fragile X premutation, autoimmune ovarian destruction, chemo/radiation, or undefined causes
- Ovarian Tumors
  - Rare Inhibin B secreting tumor --> FSH suppression --> estradiol suppression

Secondary Amenorrhea: Polyendocrine Metabolic Ovarian Syndrome (PMOS)

- Accounts for 20% of cases of amenorrhea (50% of oligomenorrhea)
- Clinical Diagnosis
- Rotterdam Criteria: must have two of the following three
  - Ovulatory dysfunction
  - Hyperandrogenism- clinical or laboratory
  - Polycystic ovaries
- Insulin resistance is very common in PMOS, but is not included in diagnostic criteria

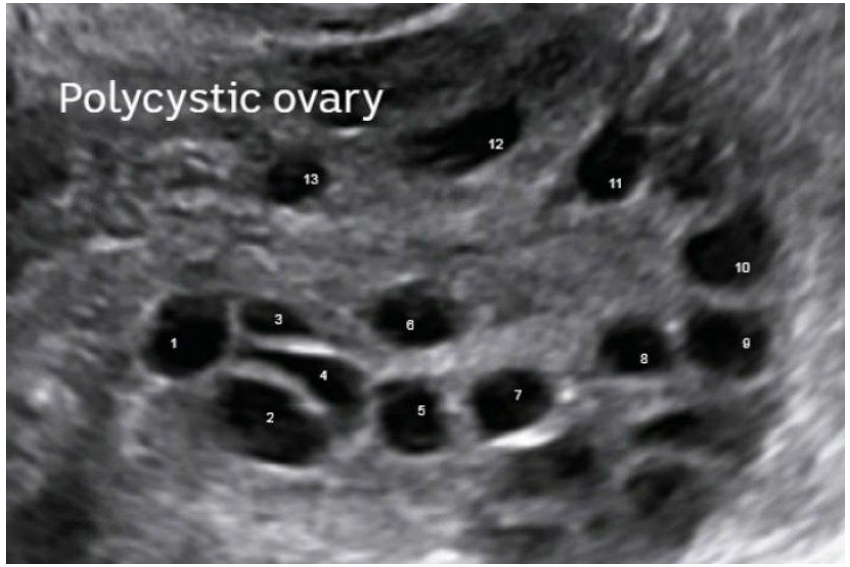
Secondary Amenorrhea: PMOS

- Hyperandrogenism
  - Total testosterone + SHBG **OR** Total + Free testosterone
    - Serum SHBG may be low → increased free testosterone
    - Low level of SHBG is a serum biomarker for insulin resistance, hyperandrogenism, and abnormal glucose and lipid metabolism
    - SHBG polymorphisms are correlated with PMOS risk
- 2-hour glucose tolerance or HgbA1C
- Fasting lipid panel
- Gonadotropin testing
  - LH increased compared to FSH (elevated LH:FSH ratio)
  - Often 2-3 x compared to approximately 1:1, but still often WNL
- Serum progesterone?

PMOS Imaging

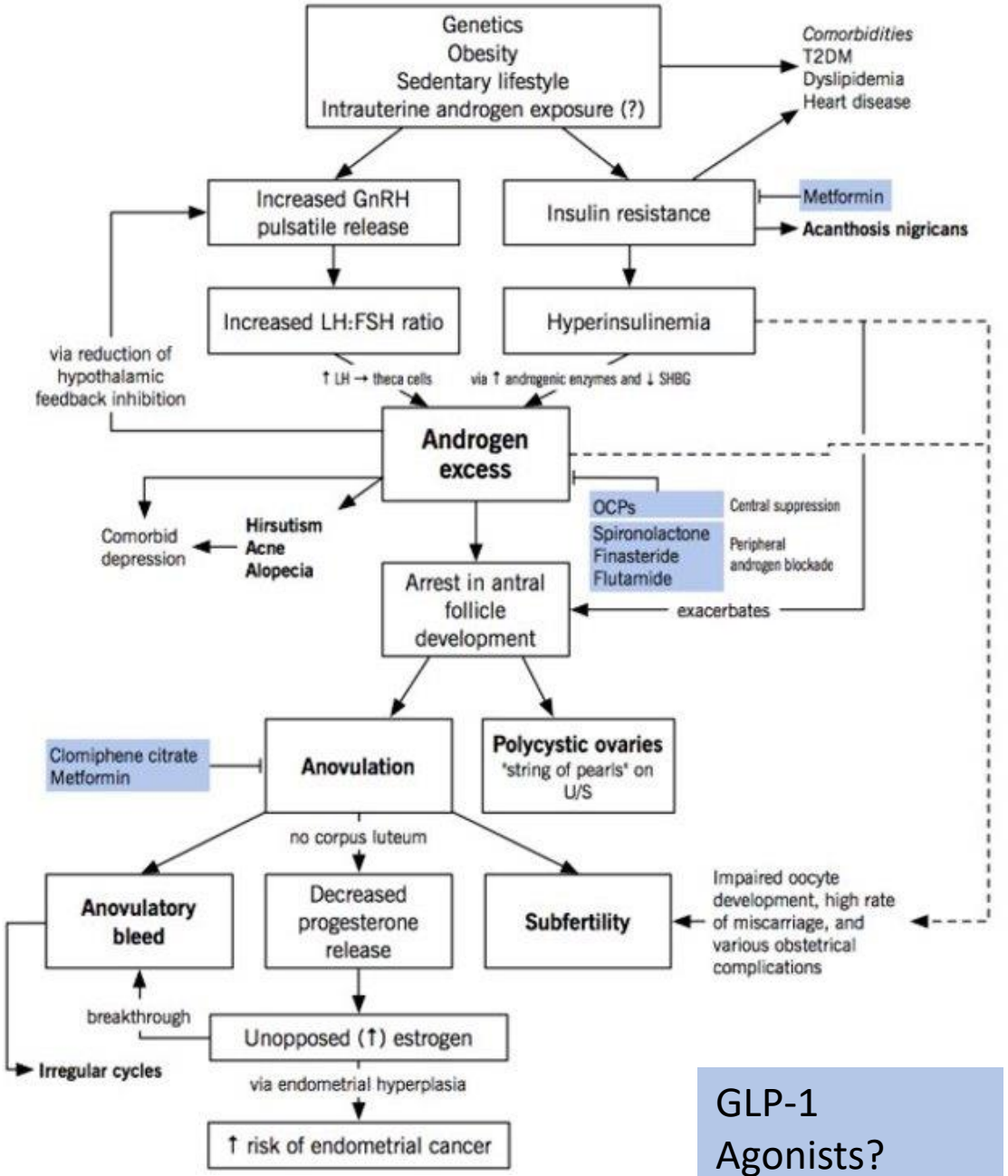
- In one or both ovaries
  - 12 or more follicles measuring 2-9mm in diameter
  - Increased ovarian volume (>10cm^3)
- Evaluate endometrial thickness

- Excessive intra-ovarian androgen causes granulosa cell dysfunction
  - Oligo-ovulation
  - Polycystic morphology: many small follicles with premature luteinization → arrest of follicle maturation, no dominant follicle



Impacts of PMOS

- Menstrual irregularity
- Acne
- Hirsutism
- Insulin resistance/diabetes
  - Weight gain
- Cardiovascular risk factors
- Infertility
- Miscarriage
- Uterine Pathology- endometrial hyperplasia
- Anxiety/depression



GLP-1 Agonists?

Secondary Amenorrhea: Other hyperandrogenism

- Non-classic/late onset congenital adrenal hyperplasia (21-hydroxylase deficiency)
  - Elevated 17-OHP
  - Milder enzyme deficiency than classic CAH
- Androgen secreting tumors
  - May cause anovulation and/or endometrial atrophy
- Exogenous androgens

Secondary Amenorrhea: Thyroid Disease

- Hypothyroidism: 10-35% will have menstrual irregularities depending on severity of thyroid dysfunction
- Hyperthyroidism: 1-4% will have menstrual irregularities depending on severity of thyroid dysfunction

- Can be heavy bleeding, abnormally light bleeding, or amenorrhea

- Primary hypothyroidism (thyroid problem) can cause pituitary cell (lactotroph and thyrotroph) hyperplasia and thus elevated prolactin in the absence of lactotroph adenoma (prolactinoma)

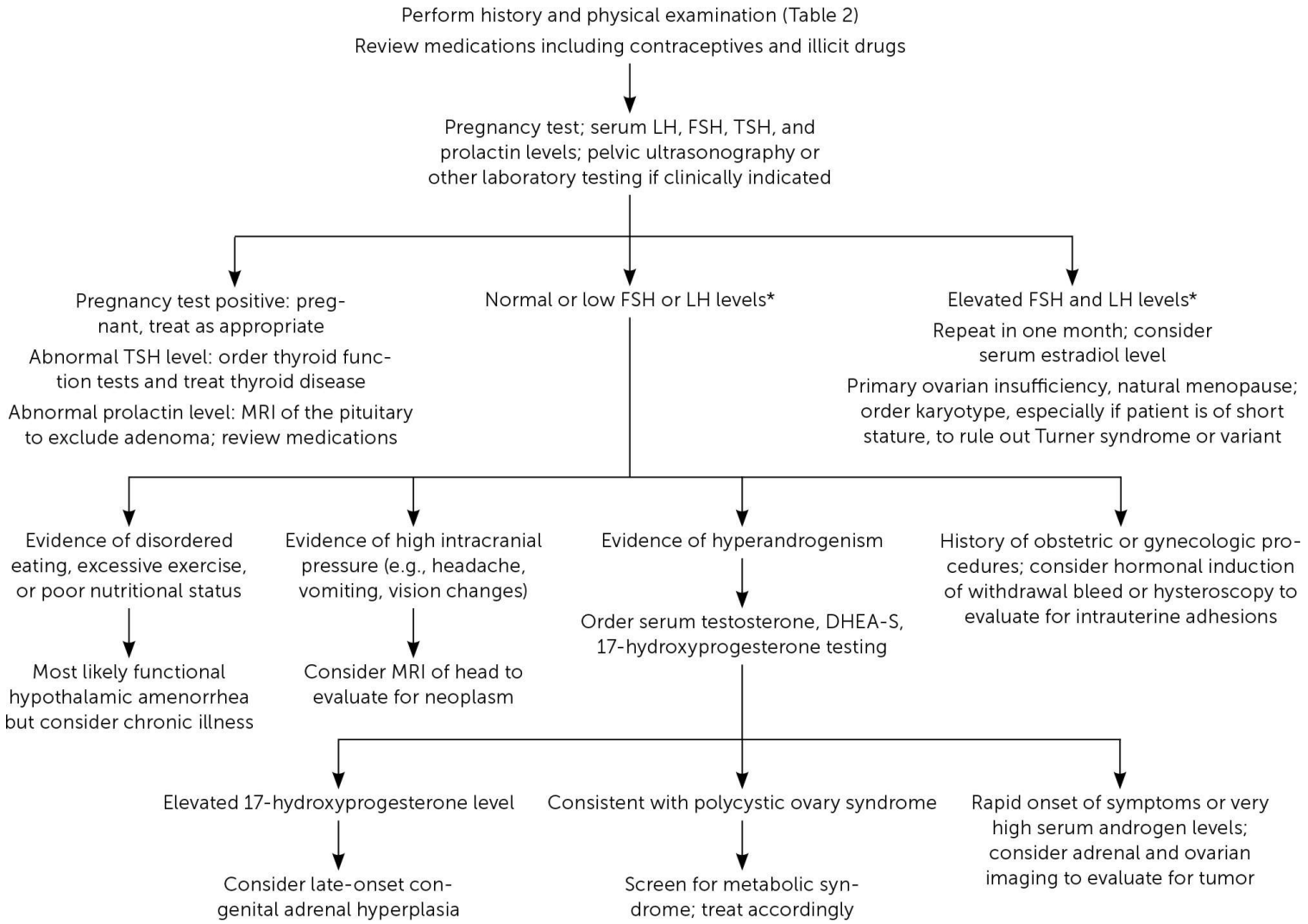
Secondary Amenorrhea: Pituitary Disease/ Dysfunction

- Lactotroph (prolactin-secreting) adenomas --> hyperprolactinemia
  - Can also be elevated by primary hypothyroidism, lactation, stress, and recent intercourse
  - Should be checked twice (ideally fasting) before imaging ordered, especially if borderline elevation
  - Elevated prolactin suppresses GnRH secretion leading to amenorrhea
- Medications: calcium-channel blockers, methyldopa, Haloperidol, Risperdal, etc can cause increase in prolactin
- Other sellar masses can cause deficient gonadotroph secretion
- Other pituitary diseases: necrosis/infarction, radiation, infiltrative lesions

Secondary Amenorrhea: Hypothalamic Dysfunction

- Functional hypothalamic amenorrhea
  - Decreased GnRH secretion --> decreased gonadotroph pulses/LH surge --> lack of ovulation
  - Eating disorders/very restricted fat intake, excessive exercise, high stress event or acute illness (typically transient)
  - Low estrogen levels can lead to bone loss over time
- Hypothalamic tumors and infiltrative diseases
- Systemic illness (T1DM, celiac disease)

Secondary Amenorrhea: Work-Up



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Secondary Amenorrhea: H&P

Findings in the Evaluation of Amenorrhea

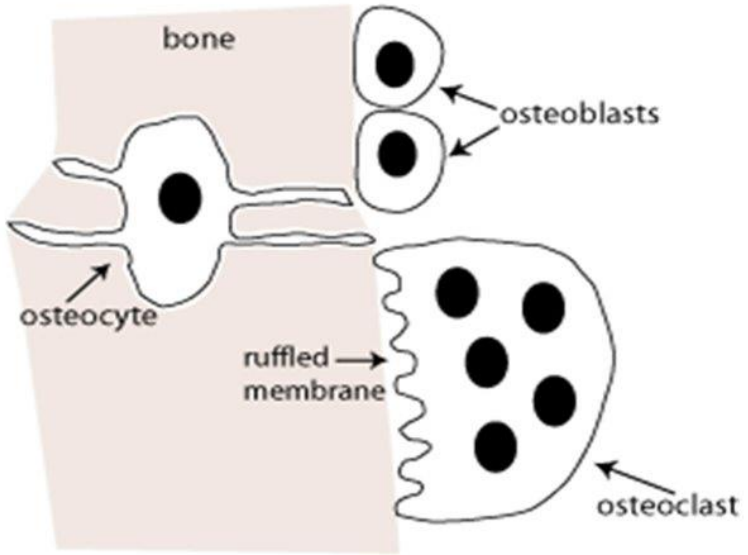
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Secondary Amenorrhea: Treatment

- Amenorrhea is a symptom, not a disease
  - Treat the underlying issue
    - Chronic disease
    - Eating disorder
    - Thyroid disease
    - Insulin resistance
- Periods themselves are not always the goal

Secondary Amenorrhea: Treatment

- If due to anovulation
  - Endometrial protection
  - Testing for other comorbidities (hyperlipidemia, diabetes)
- If due to inadequate estrogen production·
  - Correct underlying problem, if possible
  - Assess effects on other organs
    - Breast
    - Uterus
    - Bone
    - Cardiovascular health
    - Neurocognitive development



Secondary Amenorrhea: Treatment

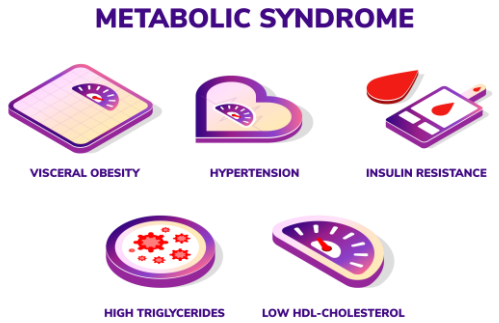
- What are your patient's concerns/goals?
  - Pregnancy?
  - Regular menses?
  - Avoiding ”hormones”?
  - Improving health concerns (diabetes, hyperlipidemia)?
  - Fixing acne and/or hirsutism?





What are my goals?

- Avoiding or managing downstream/concurrent health effects (endometrial hyperplasia, metabolic syndrome, etc)



Questions??